Alexa (Lex) Schultz

EDUCATION

Brown University, Providence, RI
Ph.D., Earth, Environmental, and Planetary Sciences ` September 2025-now

Yale University, Yale College, New Haven, CT
B.S., Earth and Planetary Sciences (Distinction in Major); 3.78 GPA ` May 2024
Thesis: *A Geochemical Model to Constrain Core-Mantle Equilibration During Large-Scale Impact-Driven Earth Accretion*

Relevant Coursework: Aqueous Geochemistry, Introduction to Geodynamics, Physical Oceanography, Mineralogy, History of Life, Introduction to Earth and Planetary Physics, Rock Formation in Mountain Belts, Earth Surface Processes, Global Tectonics, Linear Algebra, Differential Equations, Multivariable Calculus

RESEARCH EXPERIENCE

Brown University, Department of Earth, Environmental, and Planetary Sciences
PhD student September 2025 – Present
Advisor: Chris Huber & Sam Birch
Modeling the evolution of stresses in icy moons to investigate fracture propagation and possibilities of material exchange between the ocean, ice shell, and surface.

Dartmouth College, Thayer School of Engineering, Ice Research Laboratory
Research Assistant June 2024 – September 2024
Advisor: Jacob Buffo
Assisted in laboratory work, such as ice grain size analysis and heat probe testing, of ice cores, in the Ice Research Lab at Dartmouth College. Analyzed geochemical data from lab-grown ice cores and utilize computer software PHREEQC to predict habitability-relevant properties and salt precipitation during the process of freezing.

Yale University, Department of Earth and Planetary Sciences
Senior Thesis September 2023 – May 2024
Advisor: Jun Korenaga
Created a geochemical model in MATLAB to calculate core-mantle equilibration throughout impact-driven Earth accretion. Computes metal-silicate chemical equilibrium for metallic components falling through an impact-generated magma ocean. Investigated a range of planetary collision scenarios and the impact of evolving oxygen fugacity on degree of equilibration between impactor core and proto-Earth.

Yale University, Department of Earth and Planetary Sciences
Research Assistant September 2021 – May 2024
Advisor: Maureen Long
Used receiver functions to analyze the crustal structure of the Cascadia Subduction Zone forearc, with the purpose of constraining serpentinization of the mantle wedge. Downloaded seismic data from hundreds of seismic stations in the Pacific Northwest via Funclab in MATLAB, calculates receiver functions for each earthquake at each station, and manually screened for an “inverted Moho” (a velocity decrease when moving from the crust to the mantle).

Lunar and Planetary Institute Summer Intern Program
Research Intern June 2023 – August 2023
Advisor: Pat McGovern
Modeled the evolution of large volcano-tectonic structures on Venus, considering flexural loading due to magma ascent and distribution. Used MATLAB to model the growth of volcanic edifice shapes and then quantitatively characterized the topography of said model shapes. Analyzed the topographic signatures of 49 volcano-tectonic features on Venus and compared them to the topographic signatures of the models we created. Created a non-dimensional parameter—the "annularity index"—of Venus volcanic features.

Brown University, Department of Earth, Environmental, and Planetary Sciences
NSF-REU Leadership Alliance Research Intern June 2022– August 2022
Advisor: Colleen Dalton
Imaged the crust and upper mantle of the Pacific Northwest using Rayleigh wave phase velocities. Wrote MATLAB scripts to calculate phase velocities and used Generic Mapping Tool to create composite phase velocity tomography maps. Used parameter search to find values of Moho and shear velocity in the lithosphere and created GMT maps from the data. Gave an oral presentation at the Leadership Alliance National Symposium and presented a poster at Brown Summer Research Symposium.

FELLOWSHIPS

National Science Foundation Graduate Research Fellowship, 2026 Awardee

PUBLICATIONS

I. Manuscripts in Preparation

Schultz, A., Luo, Y., Long, M. D., 2025. Moho impedance contrasts throughout the Cascadia subduction zone: Evidence for mantle wedge serpentinization. In preparation for *Geophysical Research Letters.*

CONFERENCES PRESENTATIONS

Schultz A.*, Dalton C., July 2022. Cascadia's Crust: Imaging the Crust and Upper Mantle of the Pacific Northwest Using Rayleigh Wave Phase Velocity. Leadership Alliance National Symposium, Hartford, CT (oral presentation).

Schultz A.*, Dalton C., December 2022. Imaging the Crust and Upper Mantle of the Pacific Northwest Using Rayleigh Wave Phase Velocity. American Geophysical Union Fall Meeting, Chicago, IL (poster).

Schultz A.*, Luo Y., Long M., December 2023. Mapping the Moho impedance contrast throughout the Cascadia subduction zone using a receiver function approach. American Geophysical Union Fall Meeting, San Francisco, CA, (poster).

Schultz, A.*, McGovern, P., March 2024. Topographic Characterization and Evolutionary Modeling of Large Volcano-Tectonic Structures on Venus. Lunar and Planetary Science Conference, Houston, TX, (poster).

McGovern P.*, Nguyen A., **Schultz A.**, Hassan R., December 2024. From Pluto to Venus: PyGMT as a Portal to Planetary Science in the LPI Summer Intern Program. American Geophysical Union Fall Meeting, Washington, D.C. (poster)

Schultz, A.*, Murdza, A., Tomlinson, T.C., Buffo, J.J., March 2025. Brine Evolution and Habitability in Ice Shells on Ocean Worlds: Insights from Experimental Freezing and Geochemical Modeling. Lunar and Planetary Science Conference, Houston, TX, (oral presentation).

LEADERSHIP/SERVICE

Yale EPS Inclusion, Diversity, Equity, and Anti-Racism/Discrimination Committee
Member, September 2022 – May 2024
Meet every other week to address committee responsibilities in relation to diversity, equity, inclusion, and anti-discrimination.

Native and Indigenous Students Association at Yale
Co-Chair, November 2021 – November 2022
Lead Yale's Indigenous political and cultural organization; Oversaw the planning of the Henry Roe Cloud Conference, Indigenous People's Day programming, the Missing and Murdered Indigenous Women's vigil, and the biannual Yale Powwow. Assisted in the revitalization of the Yale chapter of the American Indian Science and Engineering Society.

MEMBERSHIP

American Geophysical Union (2022–Present)
American Indian Science and Engineering Society, Member (2022–Present)

TECHNICAL SKILLS

Computational: MATLAB, Generic Mapping Tool (GMT), Python
Languages: Cherokee (intermediate), Spanish (intermediate)